

FeCode_updated2025_20260310

2026-03-12

Things that need to be changed

```
Date_Run = "20260310"
plates<- c("Plate7", "Plate8", "Plate9", "Plate10", "Plate11", "Plate12", "Plate13", "Plate14")
Std_plates<- c("STD")
Run_by = "Melanie Giessner and Zoe Read"#Instrument user
Script_run_by = "Zoe Read"#Code user
Project = "COMPASS_Synoptic_CB"
Experiment="2025"
Run_notes="The Fe2 diff std curve R squared is 0.9747518, which is below the 0.985 cutoff,
but Melanie suggested this is okay because this is the difference curve whihc is more sensitive.
Removed 45 uM Fe_2 B standard because it was off.
The check stds for FE3 60 uM were all bad, and other check standards were bad on some plates.
<60% of Spikes are within range. We tend to have issues with the spikes. "#any notes from run

#Stds that should be excluded
stds_to_remove<-c("45 uM Fe_2 B")
```

File paths

```
## [1] "Tidy Data/20260310_COMPASS_SynopticCB_Fe_STD.csv"

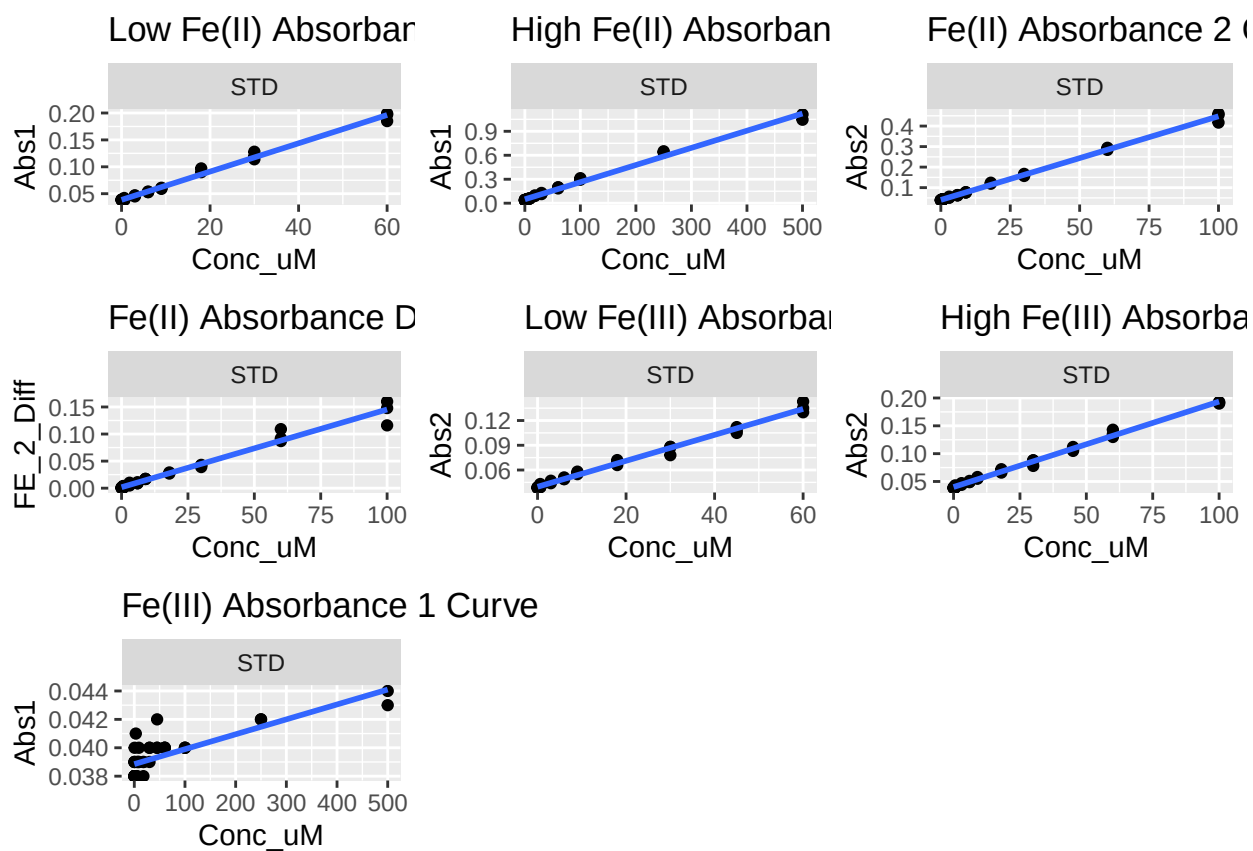
## [1] "Tidy Data/20260310_COMPASS_SynopticCB_Fe_Plate10.csv"
## [2] "Tidy Data/20260310_COMPASS_SynopticCB_Fe_Plate11.csv"
## [3] "Tidy Data/20260310_COMPASS_SynopticCB_Fe_Plate12.csv"
## [4] "Tidy Data/20260310_COMPASS_SynopticCB_Fe_Plate13.csv"
## [5] "Tidy Data/20260310_COMPASS_SynopticCB_Fe_Plate14.csv"
## [6] "Tidy Data/20260310_COMPASS_SynopticCB_Fe_Plate7.csv"
```

Plot standards

Table 1: Slopes

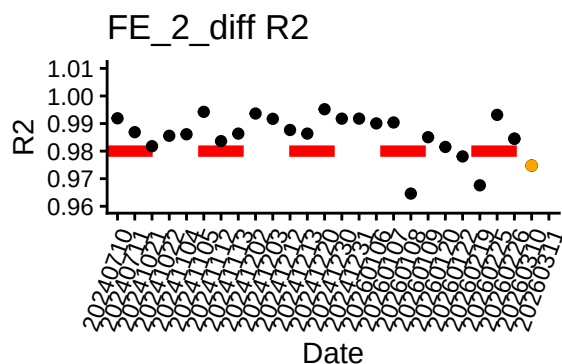
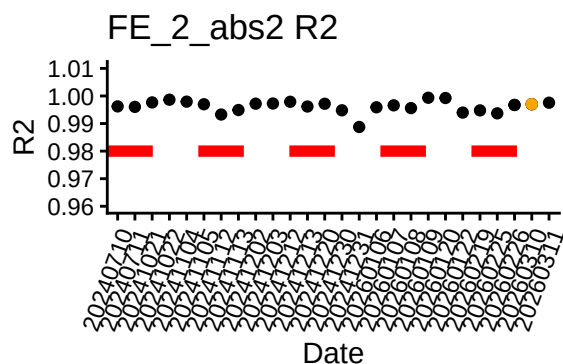
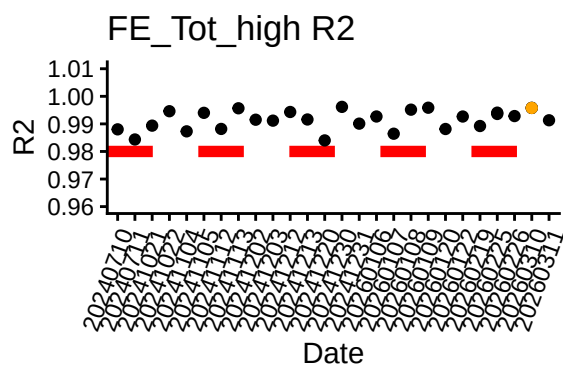
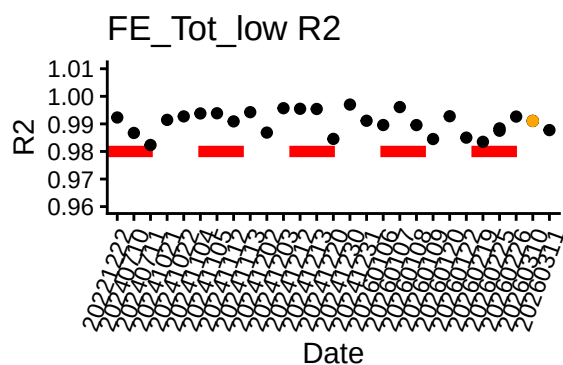
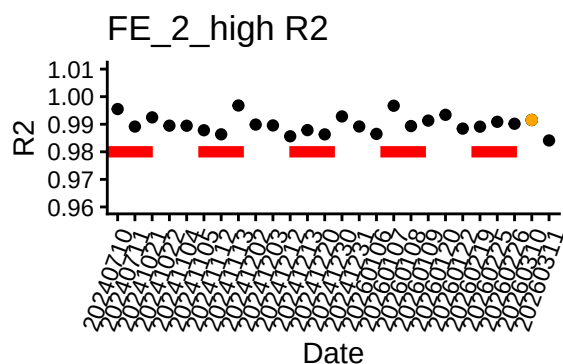
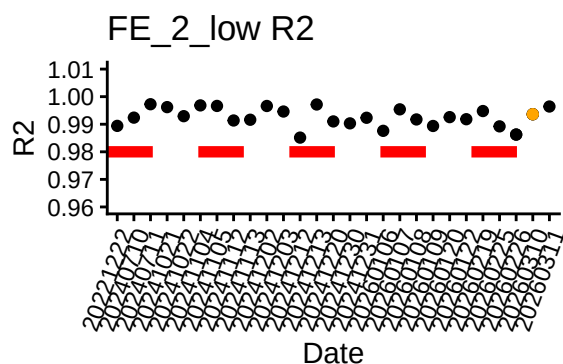
Date	Project	Curve	R2	Slope	Intercept	Top_STD
20260310	COMPASS_Synoptic_CB	Fe_2_low	0.9936312	0.0026299	0.0385411	60
20260310	COMPASS_Synoptic_CB	Fe_2_high	0.9915398	0.0021471	0.0486086	500
20260310	COMPASS_Synoptic_CB	Fe_Tot_low	0.9910689	0.0015733	0.0396655	60
20260310	COMPASS_Synoptic_CB	Fe_Tot_high	0.9957922	0.0015357	0.0400540	100

Date	Project	Curve	R2	Slope	Intercept	Top_STD
20260310	COMPASS_Synoptic_CFE_2_abs2		0.9970170	0.0040749	0.0402059	100
20260310	COMPASS_Synoptic_CFE_3_abs1		0.6610742	0.0000105	0.0388492	500
20260310	COMPASS_Synoptic_CFE_2_diff		0.9747518	0.0014364	0.0017329	100



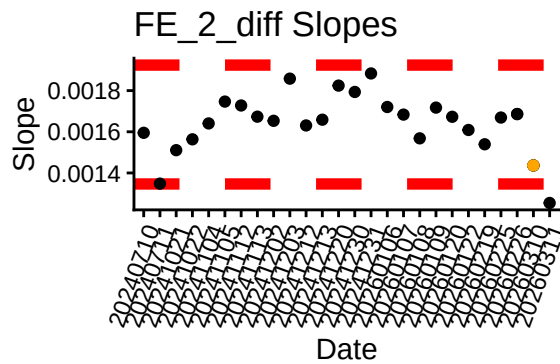
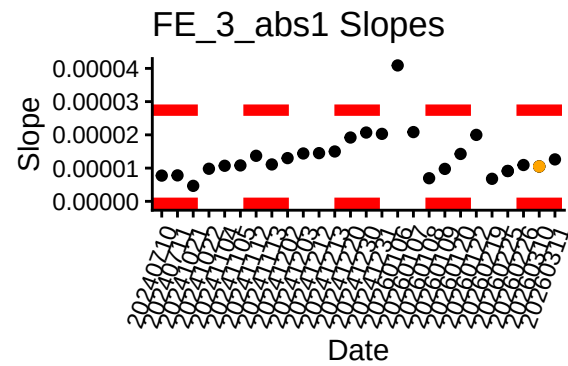
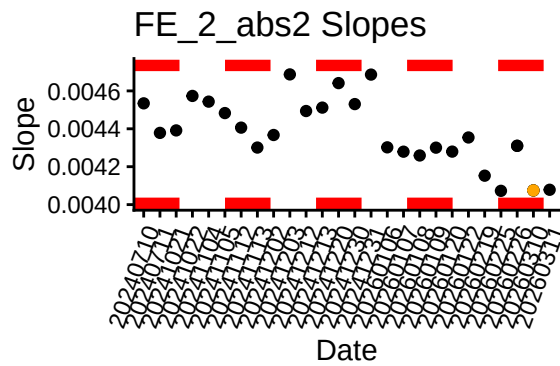
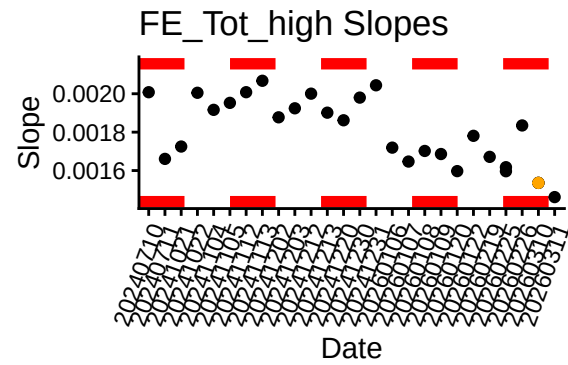
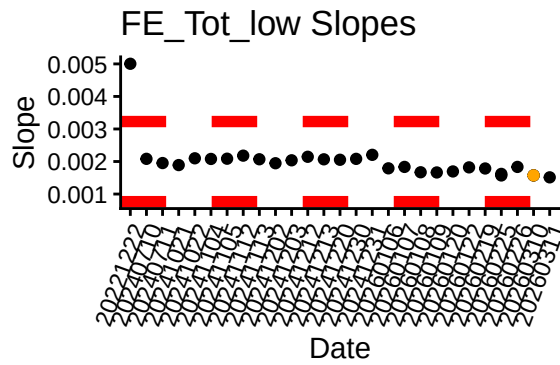
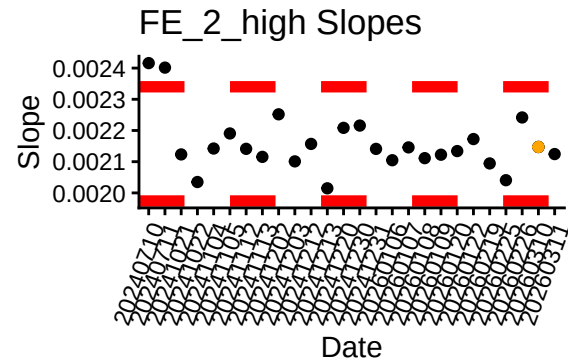
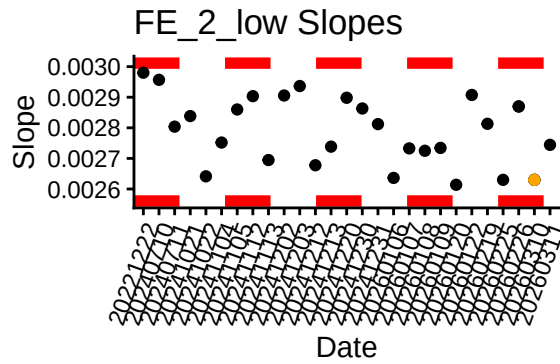
Checking STD Data against QAQC file

R2



```
## [1] "FE_2_low: Std Curve r2 GOOD"
## [1] "FE_2_high: Std Curve r2 GOOD"
## [1] "FE_Tot_low: Std Curve r2 GOOD"
## [1] "FE_Tot_high: Std Curve r2 GOOD"
## [1] "FE_2_abs2: Std Curve r2 GOOD"
## [1] "FE_3_abs1: R2 will always be low"
## [1] "FE_2_diff: Std Curve r2 is below cutoff! - REASSESS"
```

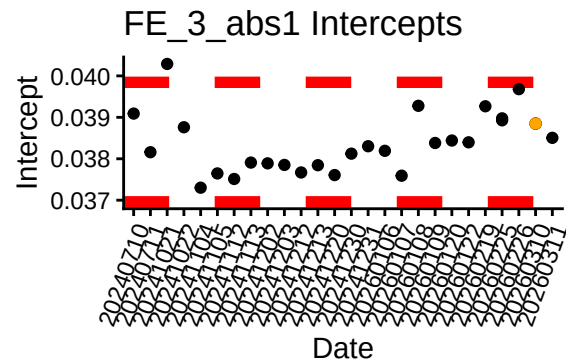
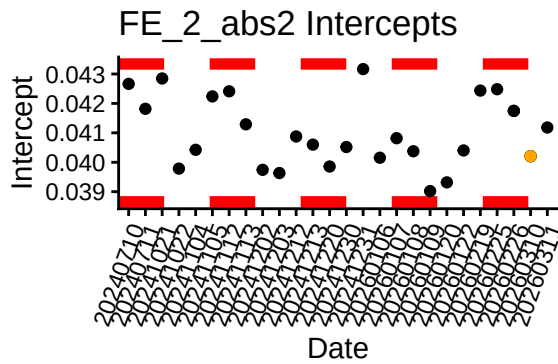
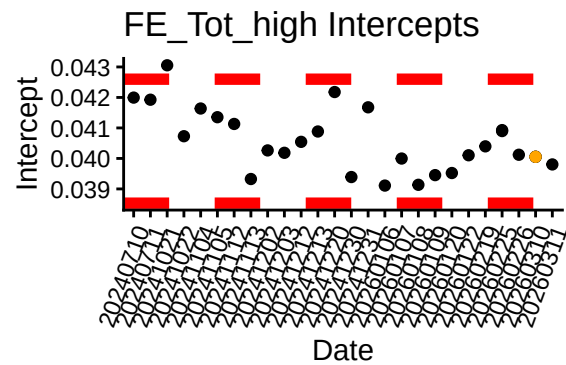
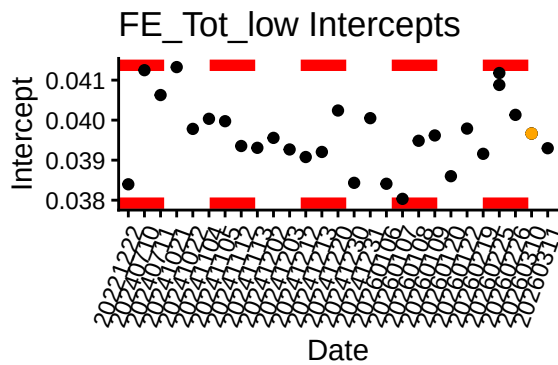
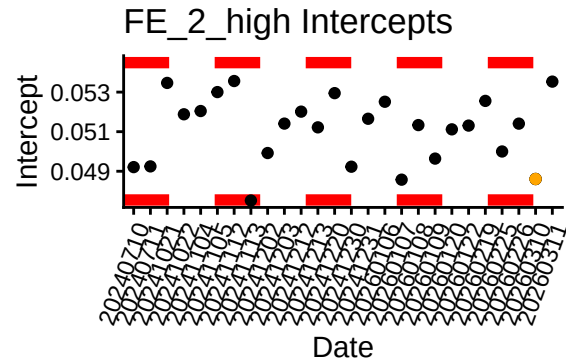
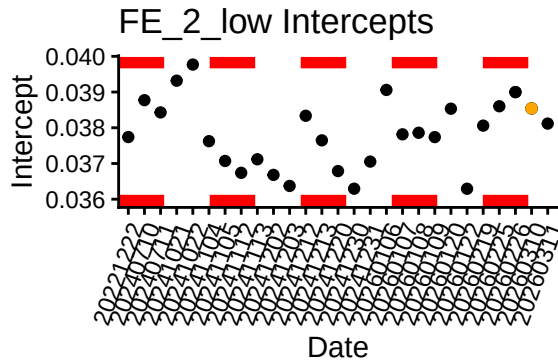
Slopes

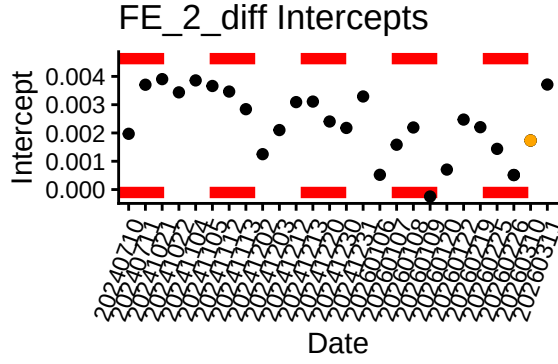


```
## [1] "FE_2_low:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_2_high:Std curve slope is with 2 sd of previous slopes"
```

```
## [1] "FE_Tot_low:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_Tot_high:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_2_abs2:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_3_abs1:Std curve slope is with 2 sd of previous slopes"
## [1] "FE_2_diff:Std curve slope is with 2 sd of previous slopes"
```

Intercepts



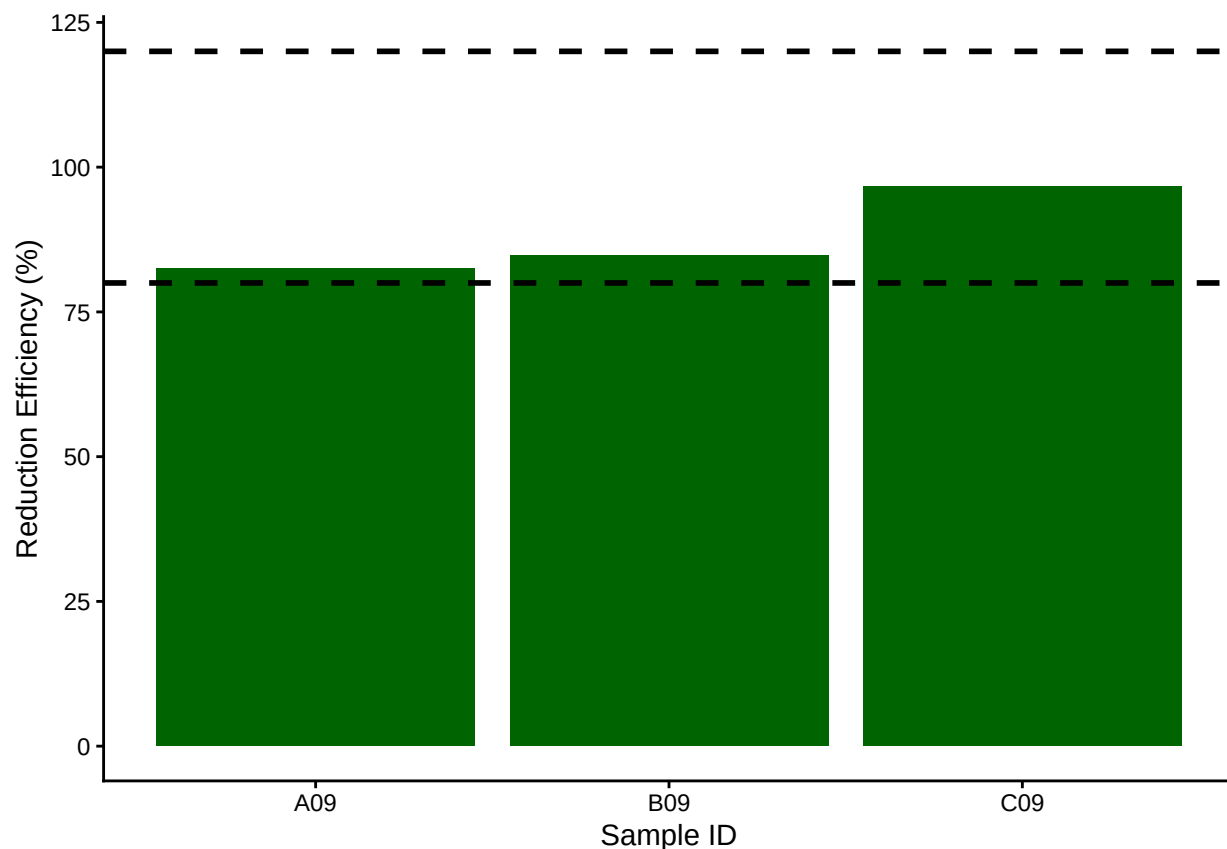


```
## [1] "FE_2_low:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_2_high:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_Tot_low:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_Tot_high:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_2_abs2:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_3_abs1:Std curve intercept is with 2 sd of previous intercepts"
## [1] "FE_2_diff:Std curve intercept is with 2 sd of previous intercepts"
```

Calculate Reduction Efficiency

Table 2: Reduction Efficiency

ID	FE2	FE3	Eff	Eff1	Eff_flag
A09	0.197	0.130	65.98985	82.48731	OK
B09	0.199	0.135	67.83920	84.79900	OK
C09	0.185	0.143	77.29730	96.62162	OK



```
## [1] ">60% of Reduction Efficiencies are within range"
```

Method Minimum Detection Limit (MDL) Calculation

I use the lowest standard to calculate MDL How to calculate MDL https://www.epa.gov/sites/default/files/2016-12/documents/mdl-procedure_rev2_12-13-2016.pdf Look at Table 1 on page 5 The Student's t-value used to calculate the method detection limit (MDL) is the one appropriate for a 99% confidence level and a standard deviation estimate with n-1 degrees of freedom

```
## [1] 0.002447186
```

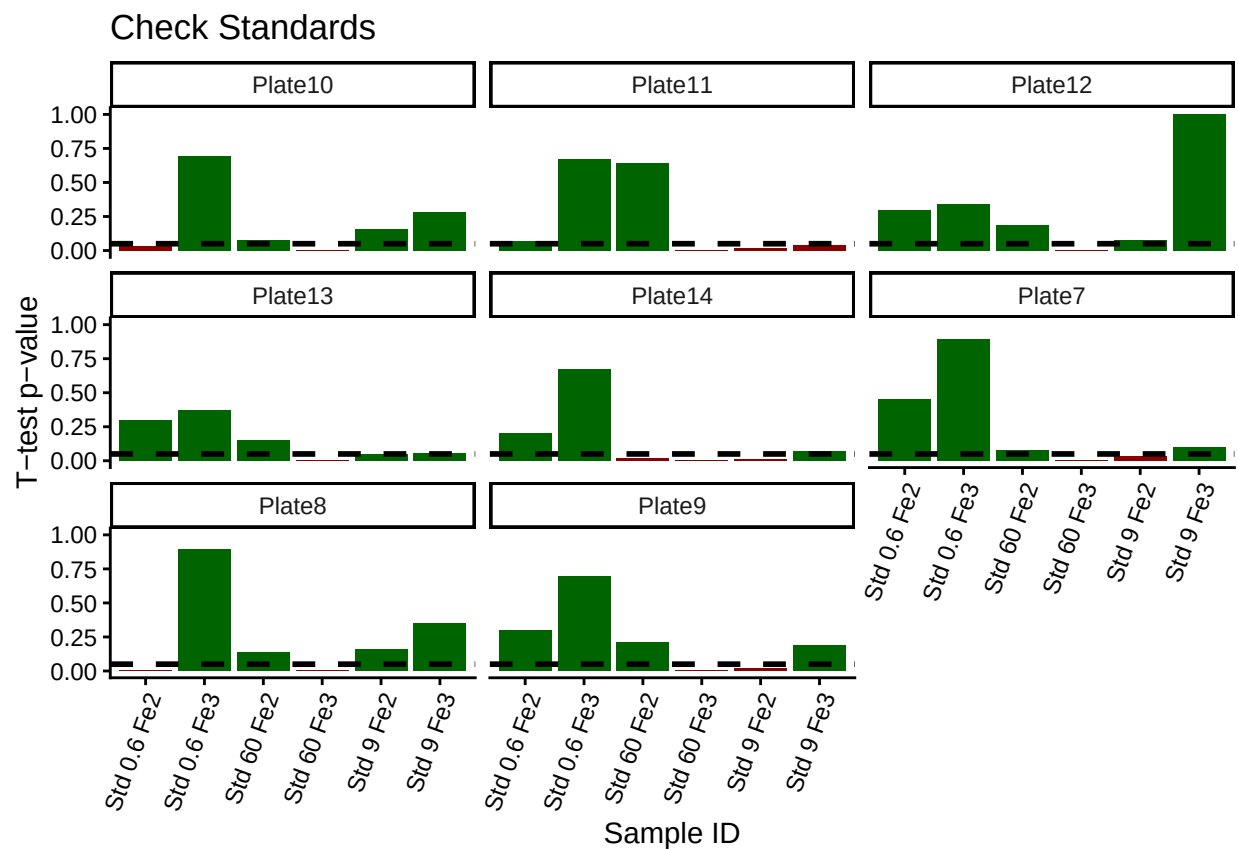
Check standards QAQC

```
## [1] "Plate7"
## [1] "Fe2 Chk Std 0.6 GOOD"
## [1] "Fe2 Chk Std 9 is significantly different from Std - REASSESS"
## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 GOOD"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"
## [1] "Plate8"
## [1] "Fe2 Chk Std 0.6 is significantly different from Std - REASSESS"
## [1] "Fe2 Chk Std 9 GOOD"
```

```

## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 GOOD"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"
## [1] "Plate9"
## [1] "Fe2 Chk Std 0.6 GOOD"
## [1] "Fe2 Chk Std 9 is significantly different from Std - REASSESS"
## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 GOOD"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"
## [1] "Plate10"
## [1] "Fe2 Chk Std 0.6 is significantly different from Std - REASSESS"
## [1] "Fe2 Chk Std 9 GOOD"
## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 GOOD"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"
## [1] "Plate11"
## [1] "Fe2 Chk Std 0.6 GOOD"
## [1] "Fe2 Chk Std 9 is significantly different from Std - REASSESS"
## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 is significantly different from Std - REASSESS"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"
## [1] "Plate12"
## [1] "Fe2 Chk Std 0.6 GOOD"
## [1] "Fe2 Chk Std 9 GOOD"
## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 GOOD"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"
## [1] "Plate13"
## [1] "Fe2 Chk Std 0.6 GOOD"
## [1] "Fe2 Chk Std 9 GOOD"
## [1] "Fe2 Chk Std 60 GOOD"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 GOOD"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"
## [1] "Plate14"
## [1] "Fe2 Chk Std 0.6 GOOD"
## [1] "Fe2 Chk Std 9 is significantly different from Std - REASSESS"
## [1] "Fe2 Chk Std 60 is significantly different from Std - REASSESS"
## [1] "Fe 3 Chk Std 0.6 Fe2 GOOD"
## [1] "Fe 3 Chk Std 9 GOOD"
## [1] "Fe 3 Chk Std 60 is significantly different from Std - REASSESS"

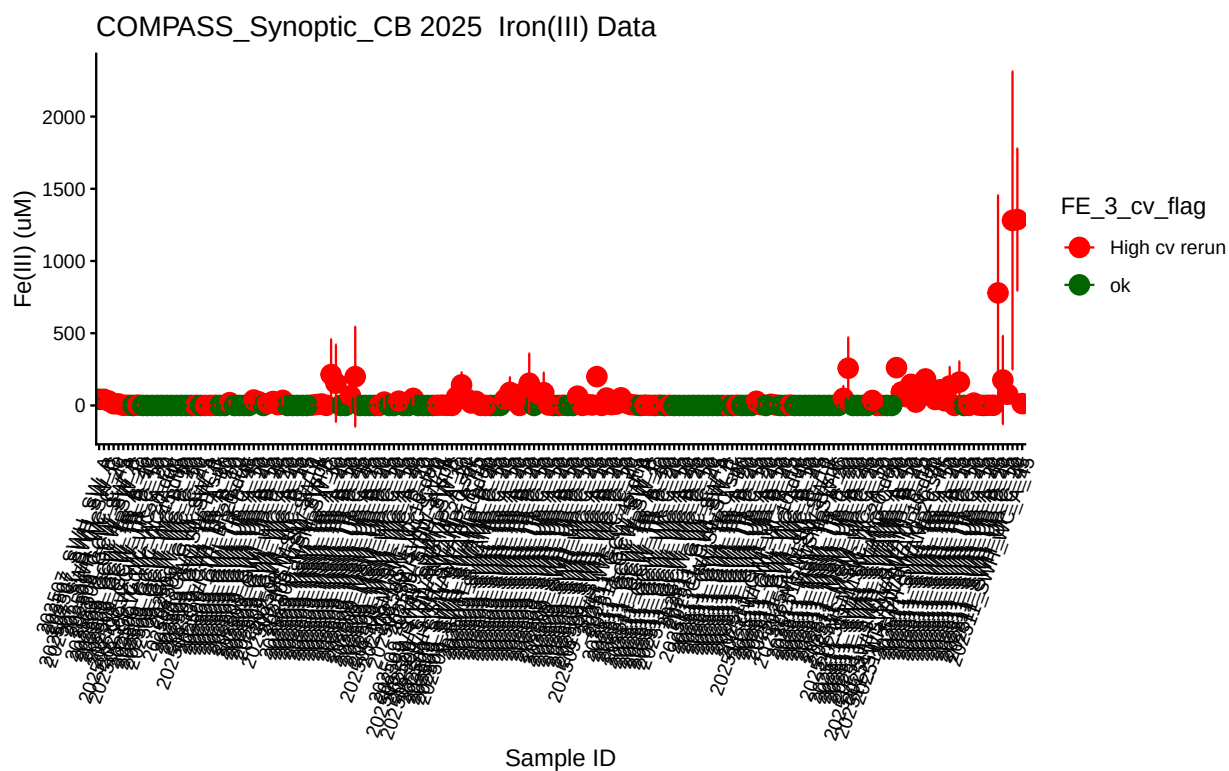
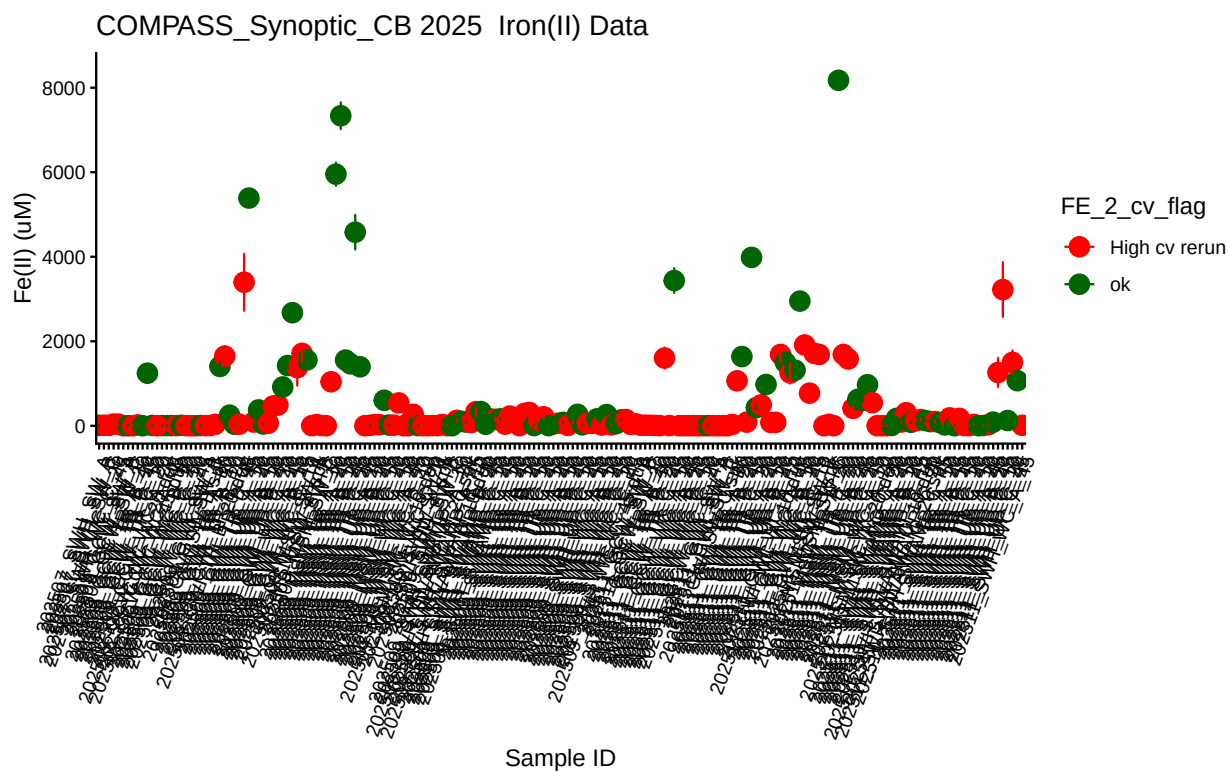
```

#No Matrix Checks We do not use matrix checks for the Fe samples because our matrix checks are not made with Trace metal grade reagents. If we want to do this in the future, we need to use trace metal grade sodium chloride and sodium bicarbonate for making the matrix checks

Subset Data and Calculate Concentrations

Calculate Averages across wells, std. dev, and cv.



Remove Bad Reps

Flagged data

Table 3: High CV Samples

ID full	FE 2 mean	FE 2 cv	FE 3 mean	FE 3 cv	Abs1 mean	Abs2 mean
Plate10_1_202511_GWI_SW_C	0.7448654	36.0972890	0.0000000	0.0000000	0.0405000	0.0440000
Plate10_8_202511_GWI_TR_C_10	430.3161395	9.7185316	43.4433706	86.8166285	0.1800000	0.2580000
Plate10_8_202511_GWI_UP_A_45	74.4036356	5.7820018	12.3275802	61.4372548	0.0630000	0.0805000
Plate10_8_202511_GWI_UP_B_10	60.7146932	3.5428159	4.4330274	126.2504637	0.0585000	0.0720000
Plate10_8_202511_GWI_WC_A_10	1368.8633952	13.4726803	0.0000000	0.0000000	0.4160000	0.5385000
Plate10_8_202511_SWH_UP_A_20	24.2108466	8.8844881	1.9143093	68.3478366	0.0465000	0.0545000
Plate11_1_202511_SWH_SWAMP_A_45	81.6863494	0.8063184	79.4736834	12.7695597	0.2240000	0.5060000
Plate11_1_202511_SWH_SWAMP_B_10	282.8848931	2.0955057	91.7529027	25.2859034	0.6560000	1.2040000
Plate11_1_202511_SWH_SWAMP_B_20	83.2388073	7.5136895	162.5648132	10.2531895	0.2273333	0.5885000
Plate11_1_202511_SWH_SWAMP_B_45	110.5620663	10.7231515	29.4778161	23.7225503	0.2860000	0.4915000
Plate11_1_202511_SWH_SWAMP_C_10 dup	121.8950090	7.5872939	152.3653177	22.7017566	0.3103333	0.7340000
Plate11_1_202511_SWH_SWAMP_C_20 spk	90.3024908	12.7642236	62.3269153	0.3080084	0.2425000	0.4690000
Plate11_1_202511_SWH_SWAMP_C_45	77.0289757	6.9728895	89.4716444	12.2361508	0.2140000	0.4620000
Plate11_1_202511_SWH_TR_A_20	226.0649340	28.5528167	201.5830140	44.7726837	0.5340000	0.8825000
Plate11_1_202511_SWH_TR_B_10	106.6032987	0.3089267	243.0192948	13.4175738	0.2775000	0.8030000
Plate11_1_202511_SWH_TR_C_10	1.5053622	17.8612310	1.0269143	68.1254192	0.0425000	0.0480000
Plate11_1_202511_SWH_WC_A_45	9.6807029	5.5548905	15.5887602	20.5855600	0.0640000	0.1040000
Plate11_8_202511_SWH_UP_C_10	125.1034225	6.1193369	89.2883795	22.8583622	0.0796667	0.1195000
Plate11_8_202511_SWH_UP_C_20	1663.2094128	0.6336204	1732.4899921	54.8001786	0.4950000	1.0180000
Plate11_8_202511_SWH_UP_C_45	1067.0655796	0.3491724	1529.0836678	23.6909869	0.3350000	0.8190000
Plate12_1_202507_SWH_WC_C_20	45.9944252	9.9379344	18.0248697	101.0673845	0.1595000	0.2200000
Plate12_1_202507_SWH_WC_C_45	43.5228106	6.1778231	16.1460924	83.8709430	0.1530000	0.2210000
Plate12_1_202509_GCW_SW_A	2.6461074	10.1611983	0.2983869	82.8207119	0.0455000	0.0515000
Plate12_1_202509_GCW_SW_B	1.3152380	0.0000000	1.5215990	59.4721152	0.0420000	0.0480000
Plate12_8_202509_GCW_TR_A_20	18.1268722	11.8664145	0.0000000	0.0000000	0.0445000	0.0510000
Plate13_1_202509_MSM_SW_A	3.2164800	0.0000000	3.2934936	13.7381030	0.0470000	0.0585000
Plate13_1_202509_MSM_SW_B	6.6387156	0.0000000	7.8266521	17.3431834	0.0560000	0.0795000
Plate13_1_202509_MSM_SW_C	1.5053622	17.8612310	1.0564635	23.3918317	0.0425000	0.0465000
Plate13_1_202509_MSM_WC_A_45	224.4348532	7.1900493	73.3655008	32.3681650	0.5305000	0.9665000
Plate13_50_202509_MSM_TR_A_10	930.8270146	12.9985807	320.7972395	69.9266167	0.0875000	0.1260000
Plate14_1_202509_GWI_SW_A	4.3572252	12.3416261	0.0000000	0.0000000	0.0500000	0.0530000
Plate14_1_202509_GWI_SW_B	5.8782188	0.0000000	0.4114831	39.7461335	0.0540000	0.0625000
Plate14_1_202509_GWI_SW_B spk	37.4388362	10.0544448	0.0000000	0.0000000	0.1370000	0.1760000
Plate14_8_202509_GWI_UP_A_10	97.2185397	11.0627551	52.8071195	15.6084013	0.0705000	0.0915000
Plate14_8_202509_GWI_UP_B_10	54.6307188	11.8120894	20.2629352	60.0481318	0.0565000	0.0720000
Plate14_8_202509_GWI_UP_B_20	425.7531586	5.5574708	41.0420262	54.2946192	0.1785000	0.2645000
Plate14_8_202509_GWI_UP_C_20	919.2715877	3.2760811	53.0497199	42.0461810	0.2953333	0.4650000
Plate7_1_202509_SWH_SWAMP_A_20	103.2655142	3.6454671	96.0551579	42.7808374	0.2703333	0.5630000
Plate7_1_202509_SWH_SWAMP_A_20 spk	87.4304436	5.8918382	39.1615556	11.4796786	0.2363333	0.4320000
Plate7_1_202509_SWH_SWAMP_A_45	87.0423292	3.4051645	25.6459229	61.0418720	0.2355000	0.3620000
Plate7_1_202509_SWH_SWAMP_B_10	382.5526902	19.2833661	0.0000000	0.0000000	0.8700000	1.2415000
Plate7_1_202509_SWH_SWAMP_C_45	29.4536197	7.3030405	65.8686759	55.3452389	0.1160000	0.2605000
Plate7_1_202509_SWH_TR_A_10	263.0910549	17.3994214	135.7377276	72.5194813	0.6135000	0.9690000
Plate7_1_202509_SWH_TR_A_20	122.6712380	3.7584725	110.8503790	15.6279813	0.3120000	0.6590000
Plate7_1_202509_SWH_TR_B_10	238.8727117	9.7885395	97.4689877	42.5734075	0.5615000	1.0550000
Plate7_1_202509_SWH_TR_B_20	256.8036004	20.7749503	230.7295525	95.9230635	0.6000000	1.3225000
Plate7_8_202509_SWH_UP_B_45	271.1188087	1.2955886	83.6772521	26.5950996	0.1276667	0.1930000
Plate7_8_202509_SWH_UP_C_20	54.6307188	3.9373631	10.0248539	55.8284204	0.0565000	0.0700000
Plate7_8_202509_SWH_UP_C_45	60.7146932	3.5428159	4.9058133	47.1138880	0.0585000	0.0690000
Plate8_1_202509_SWH_SW_B	2.6461074	10.1611983	0.2983869	82.8207119	0.0455000	0.0515000

Plate8_1_202509_SWH_TR_C_10	138.5063085	3.8043154	97.4251024	16.2583948	0.3460000	0.6210000
Plate8_1_202509_SWH_TR_C_20	241.4342672	4.6377368	133.1241373	122.0232147	0.5670000	0.9550000
Plate8_1_202509_SWH_TR_C_45	4.3572252	0.0000000	3.5247863	12.8366235	0.0500000	0.0635000
Plate8_1_202509_SWH_WC_B_10	266.1183478	6.8142156	20.1971837	32.8174231	0.6200000	1.0715000
Plate8_1_202509_SWH_WC_B_20	23.1795211	8.1198121	8.8690208	66.4677630	0.0995000	0.1350000
Plate8_1_202509_SWH_WC_B_45	49.7335345	8.3870850	14.2265848	44.7731900	0.1693333	0.2575000
Plate8_1_202509_SWH_WC_C_10	139.4377833	4.3677590	75.3786150	50.4394266	0.3480000	0.6775000
Plate8_1_202509_SWH_WC_C_45	26.6017567	19.2041762	0.0000000	0.0000000	0.1085000	0.1830000
Plate8_1_202509_SWH_WC_C_45 dup	30.9746133	10.4166423	3.7871083	80.2373244	0.1200000	0.1720000
Plate8_8_202511_GCW_TR_B_10	5.9589234	36.0972890	0.0000000	0.0000000	0.0405000	0.0415000
Plate9_1_202511_GCW_SW_B	16.3350499	11.5220558	0.0000000	0.0000000	0.0815000	0.0860000
Plate9_1_202511_MSM_WC_A_10	1.5053622	17.8612310	0.0000000	0.0000000	0.0425000	0.0465000
Plate9_50_202511_MSM_TR_C_20	1681.8176076	6.3948961	357.8136777	48.9402805	0.1270000	0.1725000
Plate9_8_202511_MSM_UP_C_45	614.3688563	16.7244363	52.0142467	98.5780804	0.2135000	0.2805000
Plate10_8_202511_SWH_UP_A_10	9.0009106	23.8976909	0.0000000	0.0000000	0.0415000	0.0433333
Plate12_1_202509_GCW_SW_C	2.2658590	11.8664145	0.0000000	0.0000000	0.0445000	0.0476667
Plate12_8_202509_MSM_UP_B_10	12.0428978	53.5836929	0.0000000	0.0000000	0.0425000	0.0460000
Plate13_1_202509_MSM_WC_A_10	1.8856106	14.2593715	0.0000000	0.0000000	0.0435000	0.0456667
Plate13_1_202509_MSM_WC_A_20	2.2658590	11.8664145	0.0000000	0.0000000	0.0445000	0.0460000
Plate13_1_202509_MSM_WC_C_10 dup	2.6461074	10.1611983	0.0000000	0.0000000	0.0455000	0.0460000
Plate13_1_202509_MSM_WC_C_20	3.5967284	14.9511551	0.0000000	0.0000000	0.0480000	0.0490000
Plate9_1_202511_GCW_SW_A	14.2436837	15.1014993	0.0000000	0.0000000	0.0760000	0.0865000
Plate9_1_202511_MSM_SW_B spk	31.3548617	15.4354756	0.0000000	0.0000000	0.1210000	0.1703333

Table 4: Samples Above the Detection Limit that should be rerun at a higher dilution

ID full	FE 2 mean	FE 3 mean	FE tot mean	FE 2 flag	FE 3 flag
Plate10_8_202511_GWI_TR_A_10	940.38502	0.00000	940.3850	Within range	High
Plate10_8_202511_GWI_WC_A_10	1368.86340	0.00000	1368.8634	Within range	High
Plate11_1_202511_SWH_SWAMP_A_20	175.45481	276.90749	452.3623	Within range	High
Plate11_1_202511_SWH_SWAMP_A_45	81.68635	79.47368	161.1600	Within range	High
Plate11_1_202511_SWH_SWAMP_B_10	282.88489	91.75290	374.6378	Within range	High
Plate11_1_202511_SWH_SWAMP_B_20	83.23881	162.56481	245.8036	Within range	High
Plate11_1_202511_SWH_SWAMP_B_45	110.56207	29.47782	140.0399	Within range	High
Plate11_1_202511_SWH_SWAMP_C_10	164.35473	125.35823	289.7130	Within range	High
Plate11_1_202511_SWH_SWAMP_C_20	99.85011	120.54319	220.3933	Within range	High
Plate11_1_202511_SWH_SWAMP_C_45	77.02898	89.47164	166.5006	Within range	High
Plate11_1_202511_SWH_TR_A_20	226.06493	201.58301	427.6479	Within range	High
Plate11_1_202511_SWH_TR_B_10	106.60330	243.01929	349.6226	Within range	High
Plate11_8_202511_SWH_UP_B_20	1063.33968	1168.91042	2232.2501	Within range	High
Plate11_8_202511_SWH_UP_B_45	3591.36212	0.00000	3591.3621	Within range	High
Plate11_8_202511_SWH_UP_C_20	1663.20941	1732.48999	3395.6994	Within range	High
Plate11_8_202511_SWH_UP_C_45	1067.06558	1529.08367	2596.1492	Within range	High
Plate13_1_202509_MSM_WC_A_45	224.43485	73.36550	297.8004	Within range	High
Plate13_50_202509_MSM_TR_A_20	5955.02924	0.00000	5955.0292	Within range	High
Plate14_8_202509_GWI_UP_C_20	919.27159	53.04972	972.3213	Within range	High
Plate7_1_202509_SWH_SWAMP_A_10	143.16368	74.03363	217.1973	Within range	High
Plate7_1_202509_SWH_SWAMP_A_20	103.26551	96.05516	199.3207	Within range	High
Plate7_1_202509_SWH_SWAMP_B_10	382.55269	0.00000	382.5527	Within range	High
Plate7_1_202509_SWH_TR_A_10	263.09105	135.73773	398.8288	Within range	High
Plate7_1_202509_SWH_TR_A_20	122.67124	110.85038	233.5216	Within range	High
Plate7_1_202509_SWH_TR_B_10	238.87271	97.46899	336.3417	Within range	High
Plate7_1_202509_SWH_TR_B_20	256.80360	230.72955	487.5332	Within range	High
Plate8_1_202509_SWH_TR_C_10	138.50631	97.42510	235.9314	Within range	High
Plate8_1_202509_SWH_TR_C_20	241.43427	133.12414	374.5584	Within range	High
Plate8_1_202509_SWH_WC_A_10	164.27711	216.15892	380.4360	Within range	High
Plate8_1_202509_SWH_WC_B_10	266.11835	20.19718	286.3155	Within range	High

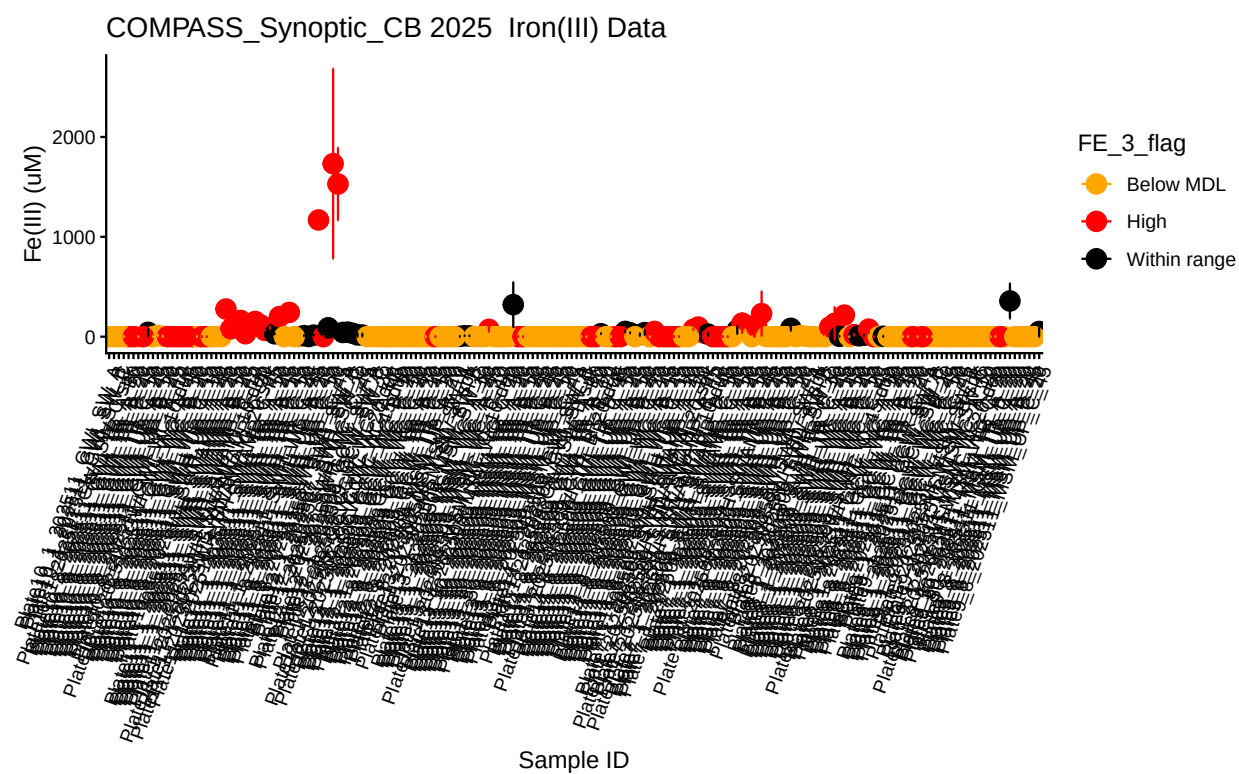
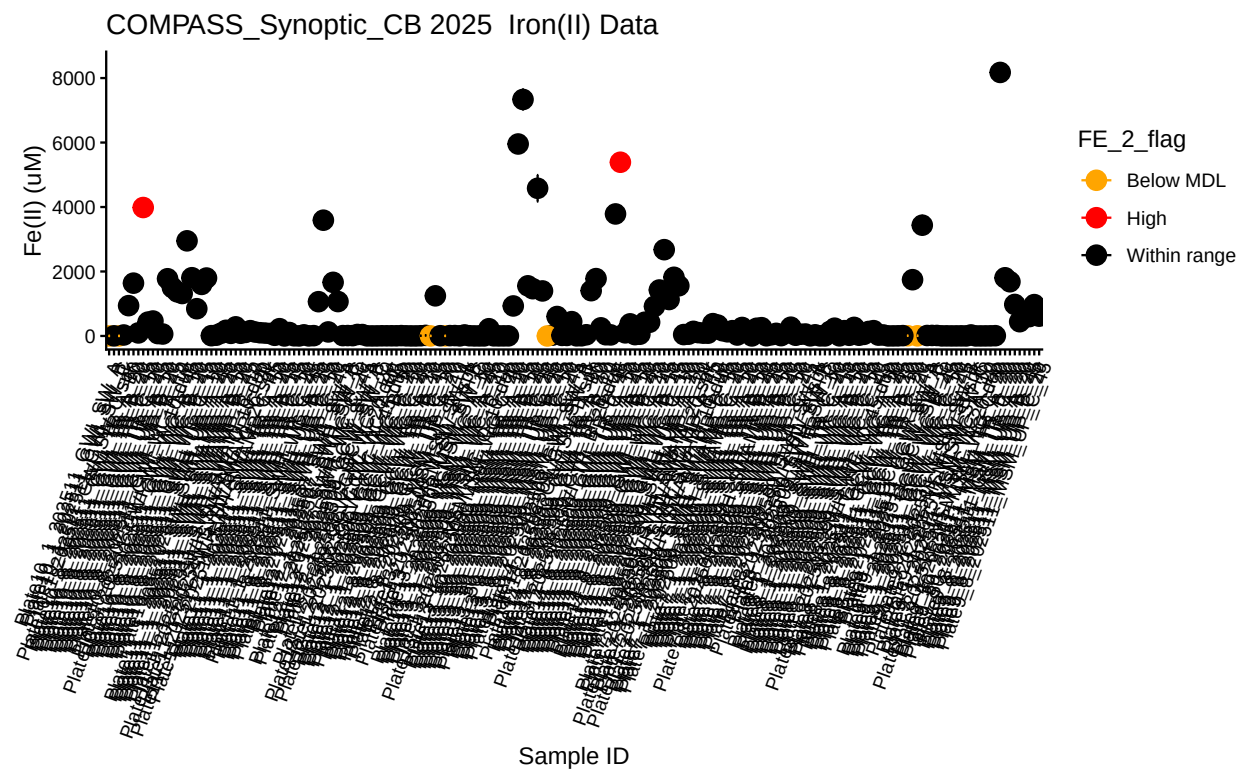
Plate8_1_202509_SWH_WC_C_10	139.43778	75.37861	214.8164	Within range	High
Plate8_1_202509_SWH_WC_C_20	164.82047	0.00000	164.8205	Within range	High
Plate10_8_202511_GWI_UP_B_20	1774.98638	0.00000	1774.9864	Within range	High
Plate10_8_202511_GWI_WC_B_10	1806.65652	0.00000	1806.6565	Within range	High
Plate10_8_202511_GWI_WC_B_45	1599.86913	0.00000	1599.8691	Within range	High
Plate10_8_202511_GWI_WC_C_10	1797.34178	0.00000	1797.3418	Within range	High
Plate14_8_202509_GWI_TR_A_20	1778.71228	0.00000	1778.7123	Within range	High
Plate14_8_202509_GWI_TR_B_45	3785.10887	0.00000	3785.1089	Within range	High
Plate14_8_202509_GWI_WC_B_10	1130.40586	0.00000	1130.4059	Within range	High
Plate14_8_202509_GWI_WC_B_45	1819.69717	0.00000	1819.6972	Within range	High
Plate7_1_202509_SWH_SWAMP_C_10	129.42443	0.00000	129.4244	Within range	High
Plate8_8_202511_GCW_TR_B_20	1745.17919	0.00000	1745.1792	Within range	High
Plate10_8_202511_GWI_TR_A_20	1638.37009	0.00000	1638.3701	Within range	High
Plate10_8_202511_GWI_TR_B_45	3984.44446	0.00000	3984.4445	High	High
Plate10_8_202511_GWI_UP_B_45	1499.26986	0.00000	1499.2699	Within range	High
Plate10_8_202511_GWI_WC_A_45	2951.12849	0.00000	2951.1285	Within range	High
Plate12_8_202509_GCW_TR_B_45	1244.66676	0.00000	1244.6668	Within range	High
Plate13_50_202509_MSM_TR_A_45	7336.71677	0.00000	7336.7168	Within range	High
Plate14_8_202509_GWI_TR_A_10	1404.88042	0.00000	1404.8804	Within range	High
Plate14_8_202509_GWI_TR_C_45	5389.10837	0.00000	5389.1084	High	High
Plate14_8_202509_GWI_WC_A_10	1425.99385	0.00000	1425.9938	Within range	High
Plate14_8_202509_GWI_WC_A_45	2676.65393	0.00000	2676.6539	Within range	High
Plate14_8_202509_GWI_WC_C_10	1557.64228	0.00000	1557.6423	Within range	High
Plate7_1_202509_SWH_SWAMP_B_45	161.48269	0.00000	161.4827	Within range	High
Plate8_8_202511_GCW_TR_C_45	3436.73732	0.00000	3436.7373	Within range	High
Plate9_50_202511_MSM_TR_A_20	8175.04403	0.00000	8175.0440	Within range	High

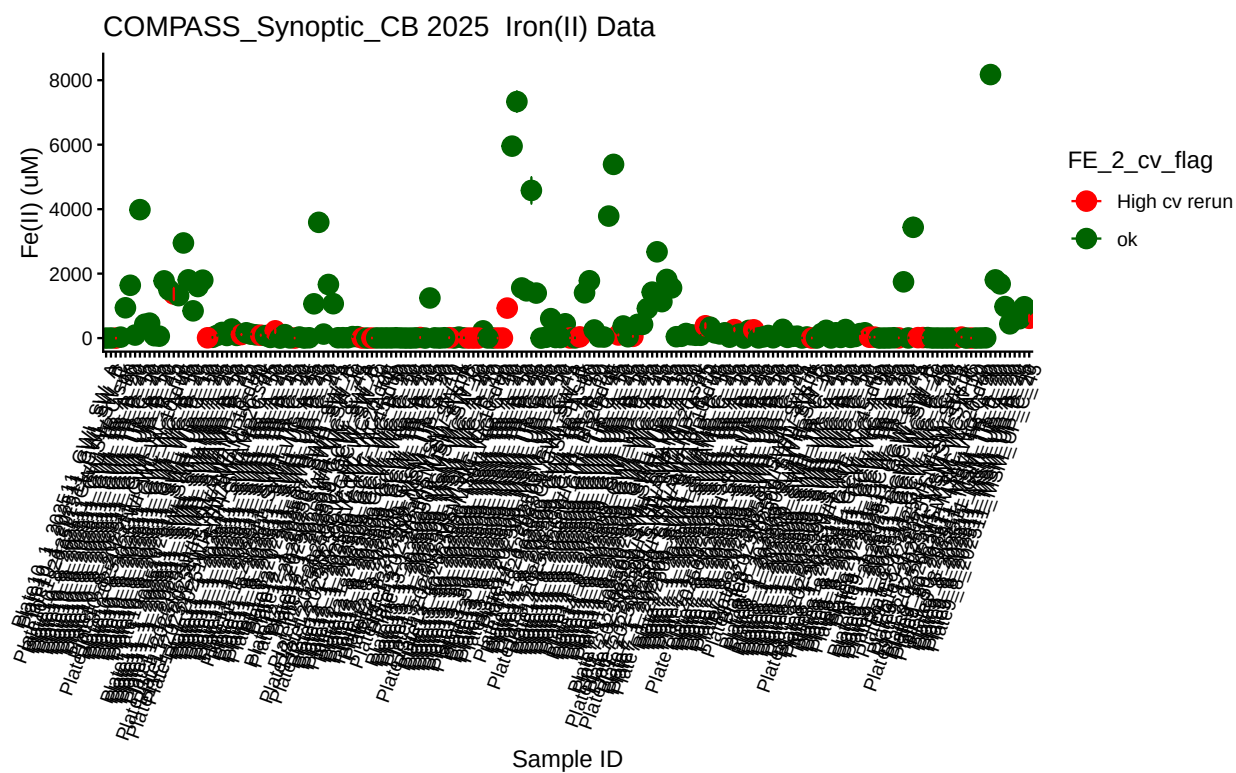
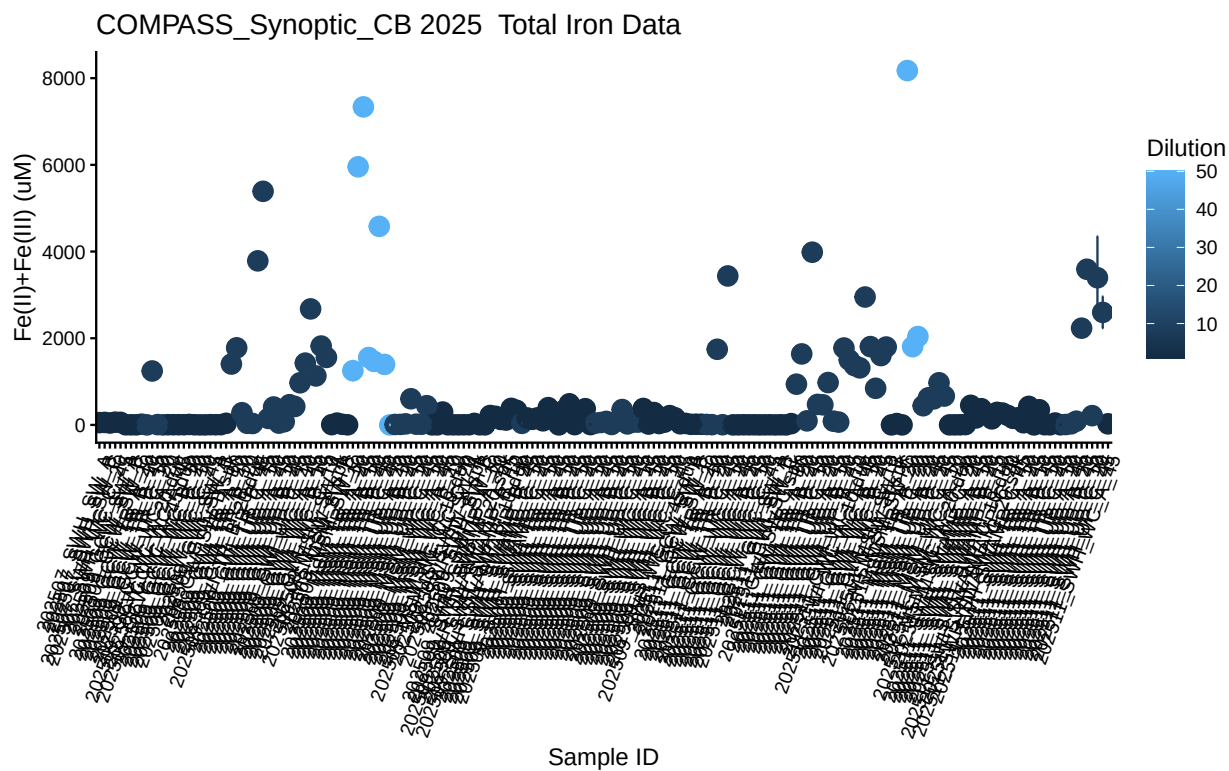
Table 5: Samples Below the Detection Limit that should be rerun at a lower dilution

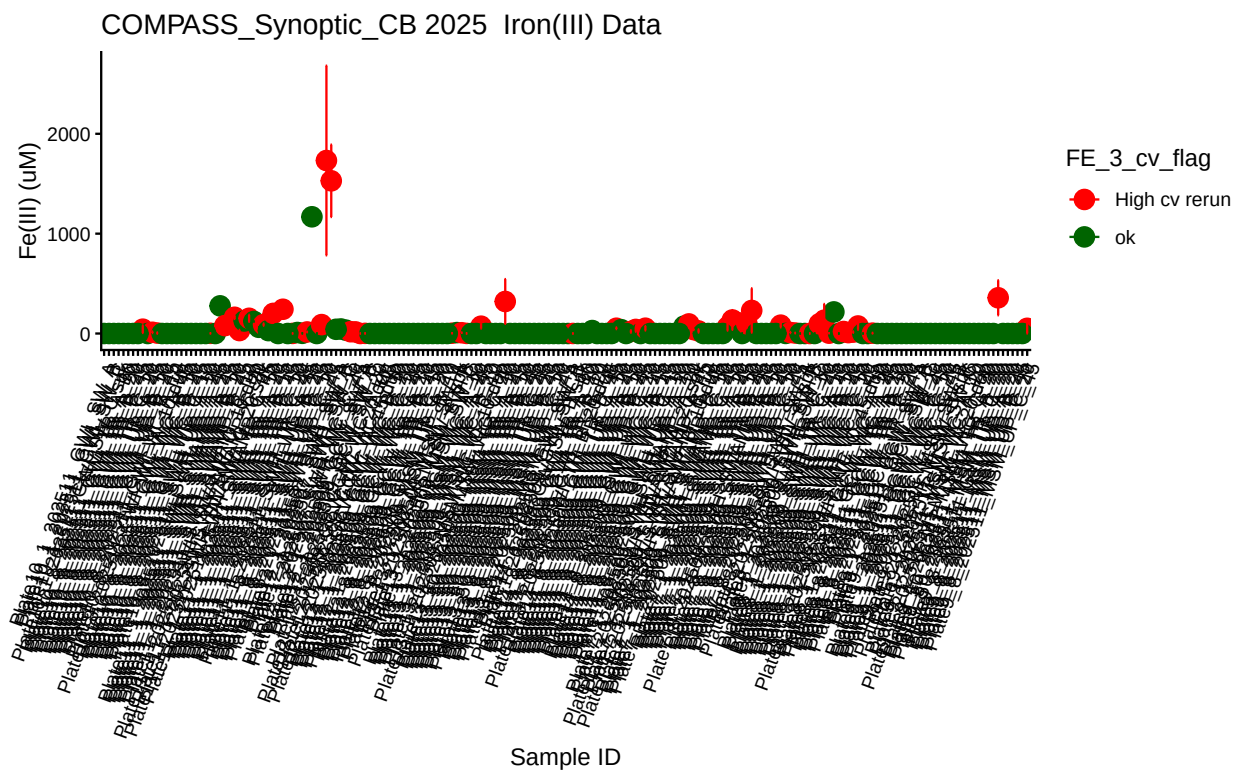
ID full	FE 2 mean	FE 3 mean	FE tot mean	FE 2 flag	FE 3 flag
Plate10_8_202511_GWI_TR_C_20	463.777999	0.0000000	463.777999	Within range	Below MDL
Plate10_8_202511_GWI_UP_A_45	74.403636	12.3275802	86.731216	Within range	Below MDL
Plate10_8_202511_GWI_UP_B_10	60.714693	4.4330274	65.147721	Within range	Below MDL
Plate10_8_202511_SWH_UP_A_20	24.210847	1.9143093	26.125156	Within range	Below MDL
Plate10_8_202511_SWH_UP_A_45	86.571585	0.0000000	86.571585	Within range	Below MDL
Plate12_8_202509_GCW_TR_A_20	18.126872	0.0000000	18.126872	Within range	Below MDL
Plate13_50_202509_MSM_TR_B_20	1466.343513	0.0000000	1466.343513	Within range	Below MDL
Plate13_50_202509_MSM_TR_B_45	4581.103999	0.0000000	4581.103999	Within range	Below MDL
Plate13_8_202509_MSM_UP_B_20	24.210847	0.0000000	24.210847	Within range	Below MDL
Plate13_8_202509_MSM_UP_B_45	603.812143	0.0000000	603.812143	Within range	Below MDL
Plate13_8_202509_MSM_UP_C_45	444.005082	0.0000000	444.005082	Within range	Below MDL
Plate14_8_202509_GWI_UP_B_45	425.753159	0.0000000	425.753159	Within range	Below MDL
Plate7_8_202509_SWH_SWAMP_B_20	36.885793	0.0000000	36.885793	Within range	Below MDL
Plate7_8_202509_SWH_UP_A_45	81.501606	0.0000000	81.501606	Within range	Below MDL
Plate7_8_202509_SWH_UP_C_10	22.689853	0.9891386	23.678992	Within range	Below MDL
Plate7_8_202509_SWH_UP_C_20	54.630719	10.0248539	64.655573	Within range	Below MDL
Plate7_8_202509_SWH_UP_C_45	60.714693	4.9058133	65.620507	Within range	Below MDL
Plate8_8_202511_GCW_TR_B_10	5.958923	0.0000000	5.958923	Below MDL	Below MDL
Plate8_8_202511_GCW_TR_C_20	4.437930	0.0000000	4.437930	Below MDL	Below MDL
Plate9_50_202511_MSM_TR_B_20	1805.398338	0.0000000	1805.398338	Within range	Below MDL
Plate10_8_202511_GWI_TR_B_20	94.176552	0.0000000	94.176552	Within range	Below MDL
Plate10_8_202511_GWI_WC_B_20	845.374592	0.0000000	845.374592	Within range	Below MDL
Plate10_8_202511_SWH_UP_A_10	9.000911	0.0000000	9.000911	Within range	Below MDL
Plate12_8_202509_GCW_TR_C_20	1.395943	0.0000000	1.395943	Below MDL	Below MDL
Plate12_8_202509_MSM_UP_B_10	12.042898	0.0000000	12.042898	Within range	Below MDL
Plate13_50_202509_MSM_TR_C_45	0.000000	0.0000000	0.000000	Below MDL	Below MDL

Plate13_8_202509_MSM_UP_C_20	16.605879	0.0000000	16.605879	Within range	Below MDL
Plate14_8_202509_SWH_UP_A_20	56.151712	0.0000000	56.151712	Within range	Below MDL
Plate7_8_202509_SWH_UP_B_10	10.521904	0.0000000	10.521904	Within range	Below MDL
Plate7_8_202509_SWH_UP_B_20	85.050591	0.0000000	85.050591	Within range	Below MDL
Plate8_8_202511_GCW_TR_A_20	10.521904	0.0000000	10.521904	Within range	Below MDL
Plate9_8_202511_MSM_UP_A_20	442.484088	0.0000000	442.484088	Within range	Below MDL
Plate12_8_202509_GCW_TR_B_10	1.395943	0.0000000	1.395943	Below MDL	Below MDL
Plate12_8_202509_GCW_TR_C_10	7.479917	0.0000000	7.479917	Within range	Below MDL
Plate12_8_202509_MSM_UP_A_20	10.521904	0.0000000	10.521904	Within range	Below MDL
Plate13_50_202509_MSM_TR_B_10	1555.068140	0.0000000	1555.068140	Within range	Below MDL
Plate13_50_202509_MSM_TR_C_20	1396.631306	0.0000000	1396.631306	Within range	Below MDL
Plate13_8_202509_MSM_UP_C_10	17.619874	0.0000000	17.619874	Within range	Below MDL
Plate14_8_202509_GWI_TR_B_20	37.899789	0.0000000	37.899789	Within range	Below MDL
Plate14_8_202509_GWI_UP_A_45	41.955772	0.0000000	41.955772	Within range	Below MDL
Plate14_8_202509_SWH_UP_A_10	42.969768	0.0000000	42.969768	Within range	Below MDL
Plate9_8_202511_GWI_UP_A_20	977.644005	0.0000000	977.644005	Within range	Below MDL
Plate9_8_202511_MSM_UP_B_20	626.167536	0.0000000	626.167536	Within range	Below MDL
Plate9_8_202511_MSM_UP_B_45	601.328210	0.0000000	601.328210	Within range	Below MDL
Plate9_8_202511_MSM_UP_C_20	972.676139	0.0000000	972.676139	Within range	Below MDL

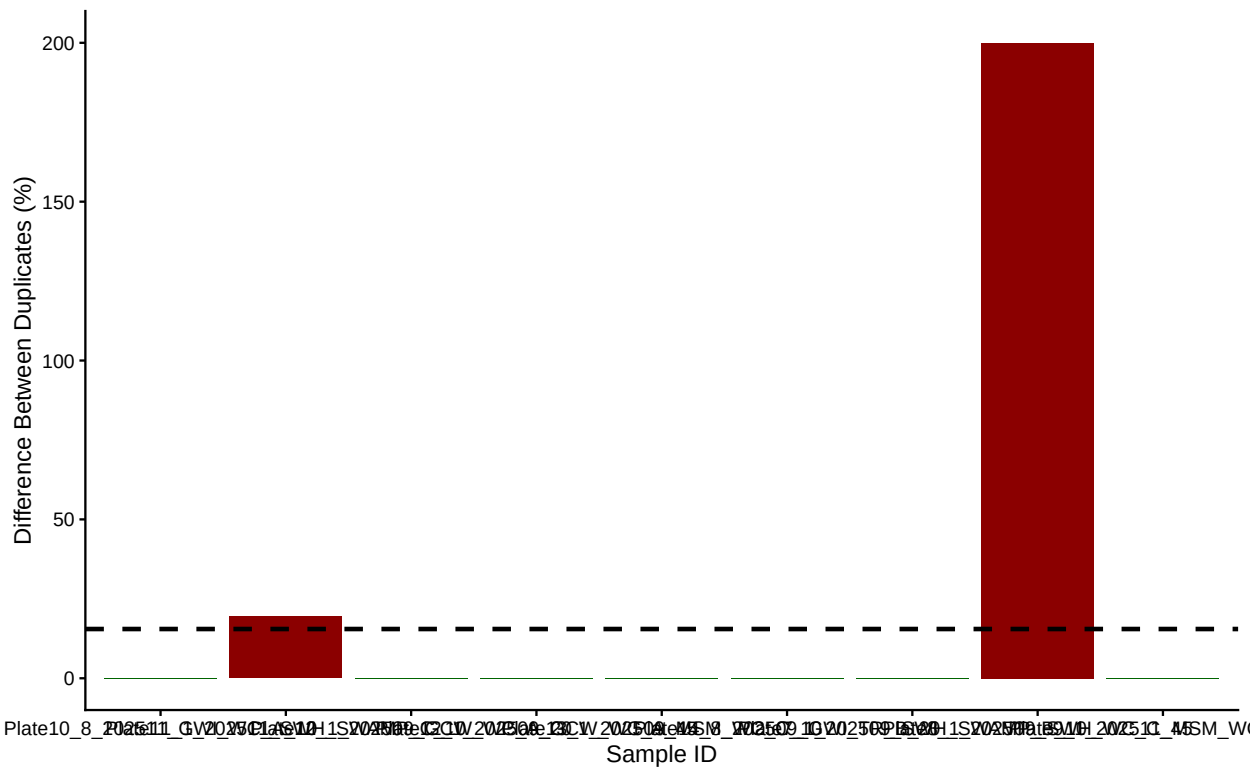
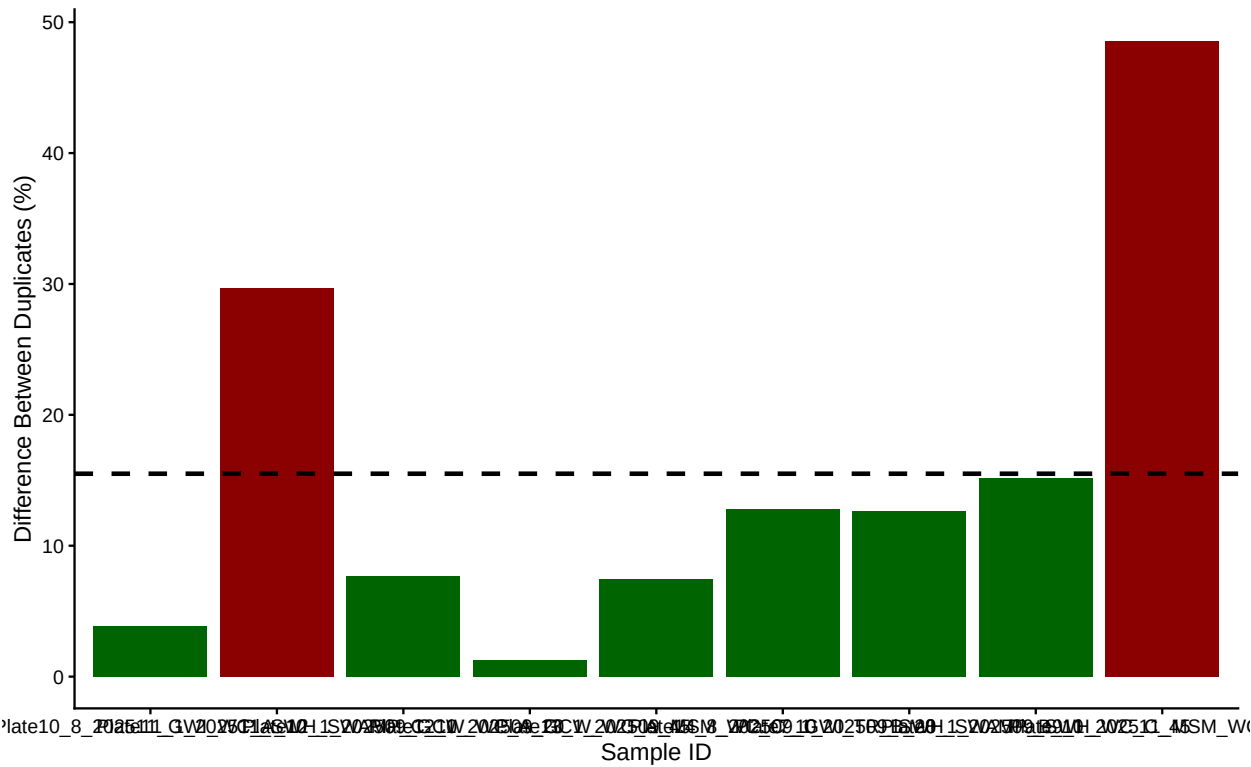
Plot data after bad reps were removed

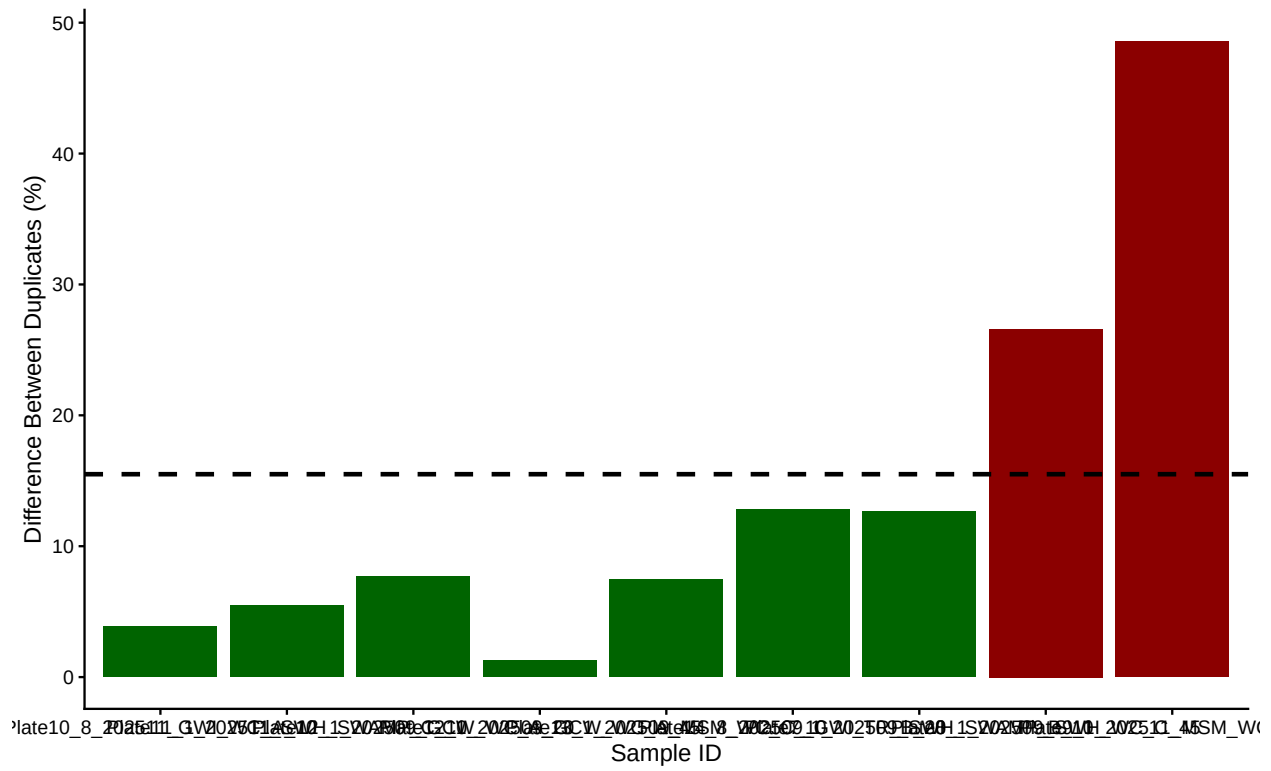






Check Dups for QAQC

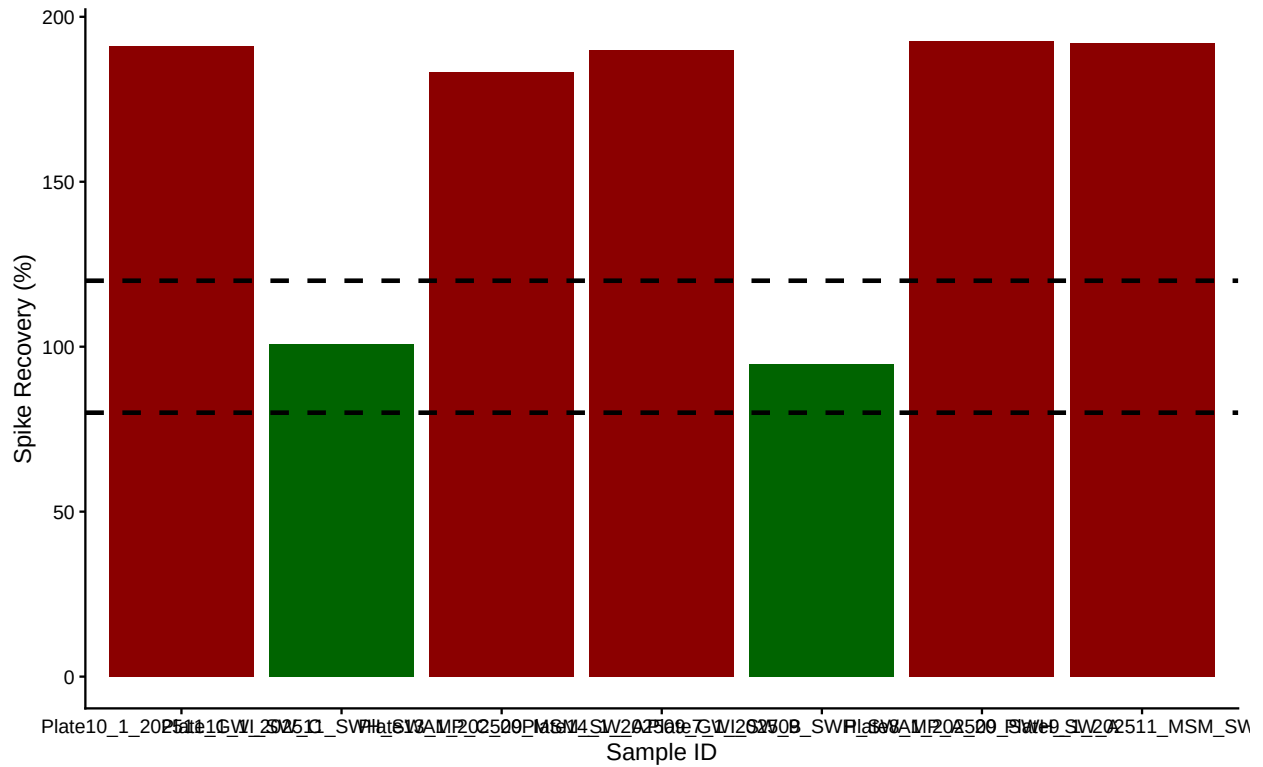




```
## [1] ">60% of Fe 2 Duplicates are within <10%"
```

```
## [1] ">60% of Fe 3 Duplicates are within <10%"
```

Check the spks for QAQC



```
## [1] "<60% of Spikes are within range - REASSESS"
```

Export full data then just final data